

Solar Fe II appendix

Sun · Fe II · downloadable appendix

Fe.json v1.163 · 15 Fe II products · 92 total

Solar Fe I anchor: 7.477 ± 0.052

HARPS · Reference Grade · Amarsi 3D-NLTE · $n = 21$. The selected Fe I product anchors this Solar appendix; products and engines are not averaged together.

Highlighted band products

near-UV

7.422 ± 0.119

Kitt Peak · Synth · 1D-LTE · Gerber · Deep Grade · $n = 12$

VIS

7.531 ± 0.117

HARPS · Synth · 1D-NLTE · Bergemann · Deep Grade · $n = 2$

Error-bar forest

How to read this forest. Each measured row is a separate abundance product for an instrument, holding, line pool and treatment; engines are never averaged together. Solid bars show statistical uncertainty from the line measurements; wireframes show systematic uncertainty from the product’s adopted error budget, which does not average away with more lines. Green and gold regions mark the Asplund and Lodders literature values and uncertainties, not pass/fail gates. Grades describe the line pool’s provenance and selection, not the quality of an engine. Reddish-orange mean-3D / Frankenstein rows are experimental and not adopted. Unmeasured model rows remain N/A.

Grades and model labels

Deep Grade. The corresponding laboratory-data pool above that depth gate. These stronger lines extend coverage but require particular care with saturation and blends.

1D-LTE uses a one-dimensional atmosphere with atomic populations in local thermodynamic equilibrium.

1D-NLTE allows departures from that equilibrium; the named implementation supplies the correction or synthesis treatment.

Synthesis fits the spectral line profile. See [Methodology](#) for the full model, line-selection and uncertainty procedures.

Asplund, Amarsi & Grevesse 2021 7.46 ± 0.04 · Lodders+ 2025 7.49 ± 0.01

blue solid = statistical σ · wireframe = systematic σ · blue = regular model result · reddish orange = experimental Frankenstein / Gerber mean-3D; not adopted · green = Asplund; gold = Lodders · N/A = no product for this model

near-UV — Kitt Peak — Kurucz 2005 corrected solar_kpno_kurucz2005_corrected · DEEPGRADED (no graded product in this band) A(Fe) dex; axis 7.26-7.78		
EW · 1D-LTE no product		N/A
Synth · 1D-LTE Deep Grade · n=12		7.617 ± 0.052 stat ± 0.110 syst
Synth · 1D-LTE · Gerber Deep Grade · n=12		7.622 ± 0.053 stat ± 0.110 syst

near-UV — Kitt Peak — Kurucz 2005 corrected solar_kpno_kurucz2005_corrected · DEEPGRADED (no graded product in this band) A(Fe) dex; axis 7.26-7.78

EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Deep Grade · n=7		7.635 ±0.081 stat ±0.110 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

near-UV — Kitt Peak — 1984 Molecfit corrected solar_kpno_molecfit_corrected · DEEPGRADED (no graded product in this band) A(Fe) dex; axis 7.26-7.78

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Deep Grade · n=12		7.402 ±0.047 stat ±0.111 syst
Synth · 1D-LTE · Gerber Deep Grade · n=12		7.422 ±0.045 stat ±0.110 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Deep Grade · n=7		7.404 ±0.051 stat ±0.110 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

near-UV axis: 7.26 · 7.39 · 7.52 · 7.65 · 7.78 A(Fe) dex

VIS — Kitt Peak — Kurucz 2005 corrected solar_kpno_kurucz2005_corrected · DEEPGRADED (no graded product in this band) A(Fe) dex; axis 7.15-8.12

EW · 1D-LTE no product		N/A
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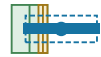
**VIS — Kitt Peak — Kurucz 2005 corrected solar_kpno_kurucz2005_corrected · DEEPGRADED (no graded product in this band)
A(Fe) dex; axis 7.15-8.12**

Synth · 1D-LTE Deep Grade · n=8		7.571 ±0.147 stat ±0.093 syst
Synth · 1D-LTE · Gerber Deep Grade · n=8		7.567 ±0.151 stat ±0.065 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Deep Grade · n=2		7.324 ±0.139 stat ±0.089 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

**VIS — Kitt Peak — 1984 Molecfit corrected solar_kpno_molecfit_corrected · DEEPGRADED (no graded product in this band)
A(Fe) dex; axis 7.15-8.12**

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Deep Grade · n=8		7.560 ±0.140 stat ±0.100 syst
Synth · 1D-LTE · Gerber Deep Grade · n=8		7.556 ±0.144 stat ±0.065 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Deep Grade · n=2		7.328 ±0.131 stat ±0.076 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

VIS — HARPS solar_harps_molecfits_corrected · DEEPGRADED (no graded product in this band)
A(Fe) dex; axis 7.15-8.12

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Deep Grade · n=8		7.909 ±0.164 stat ±0.096 syst
Synth · 1D-LTE · Gerber Deep Grade · n=8		7.925 ±0.167 stat ±0.065 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Deep Grade · n=2		7.531 ±0.086 stat ±0.080 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

VIS axis: 7.15 · 7.40 · 7.64 · 7.88 · 8.12 A(Fe) dex

Read all product values, including alternate line sets and experiments

HARPS · Synth · 1D-LTE · Deep Grade · solar_harps_molecfits_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.909 ± 0.190 reported total; n=8; ξ ALIASED

HARPS · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_harps_molecfits_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.531 ± 0.117 reported total; n=2; ξ ALIASED

HARPS · Synth · 1D-LTE · Gerber · Deep Grade · solar_harps_molecfits_corrected · DEEPGRADED · SYNTH · synth-1D-LTE-gerber: 7.925 ± 0.179 reported total; n=8; ξ NOT_IN_CAMPAIGN · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.571 ± 0.175 reported total; n=8; ξ ALIASED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.324 ± 0.165 reported total; n=2; ξ ALIASED

Kitt Peak · Synth · 1D-LTE · Gerber · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · synth-1D-LTE-gerber: 7.567 ± 0.164 reported total; n=8; ξ NOT_IN_CAMPAIGN · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_molecfits_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.560 ± 0.172 reported total; n=8; ξ ALIASED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_kpno_molecfits_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.328 ± 0.151 reported total; n=2; ξ ALIASED

Kitt Peak · Synth · 1D-LTE · Gerber · Deep Grade · solar_kpno_molecfits_corrected · DEEPGRADED · SYNTH · synth-1D-LTE-gerber: 7.556 ± 0.158 reported total; n=8; ξ NOT_IN_CAMPAIGN · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.617 ± 0.122 reported total; n=12; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.635 ± 0.137 reported total; n=7; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · synth-1D-LTE-gerber: 7.622 ± 0.122 reported total; n=12; ξ NOT_IN_CAMPAIGN · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_molecfit_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.402 ± 0.121 reported total; n=12; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_kpno_molecfit_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.404 ± 0.121 reported total; n=7; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Deep Grade · solar_kpno_molecfit_corrected · DEEPGRADED · SYNTH · synth-1D-LTE-gerber: 7.422 ± 0.119 reported total; n=12; ξ NOT_IN_CAMPAIGN · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Solar iron context

Fe II traces singly ionized iron and is retained as an ionization and consistency diagnostic alongside Fe I, not an independent headline to average with it. The current resolved pool contains strong, saturation-sensitive lines and documented blend exclusions; the diagnostics below distinguish those limitations from model and holding differences.

Near-UV opacity — report and noted

Molecular opacity INCLUDED — OH A-X, NH, CH and CN violet, 6,223 VALD lines as Turbospectrum lists with the dominant isotopologue (RYA-1207). The band previously carried none: every molecular list iSpec ships begins at 400 nm. Measured PAIRED against a molecules-off run on the same code and pool, the lever is -0.047..-0.055 dex on Fe I and -0.276..-0.332 on Fe II, with chi2r median 81.8 -> 67.4 and control windows containing no molecular line bit-identical. RYA-1209 additionally curated out Fe I 3721.272 (systematic:K07 — gf attributed to a VALD3 reference that publishes no per-line uncertainty, no NIST accuracy class, so it never met the DEEPGRADED tier's lab-gf definition); the Fe I pool is 57 candidates and the budget clears at gf rung 3 (cited lab), 55/55 GF-LAB, RMS 0.0509 dex. RESIDUAL DEFICIT: the correction still UNDER-corrects — RYA-1204 measured synth/obs above 1 on 24 of 40 windows after it — so the band remains short of opacity by an amount not yet measured on this pool, and these values are a floor rather than a converged answer. sigma_syst is unchanged at 0.1122 because it is dominated by the 0.10 dex pseudo-continuum term, not by gf: the true continuum is never observed here. Not adopted for the headline, which rests on the VIS anchor 7.466 (RYA-161: nothing calibrated to a target).

Frankenstein — experimental, pending the Bride

4 experimental products remain available in the Fe I product values and downloads.

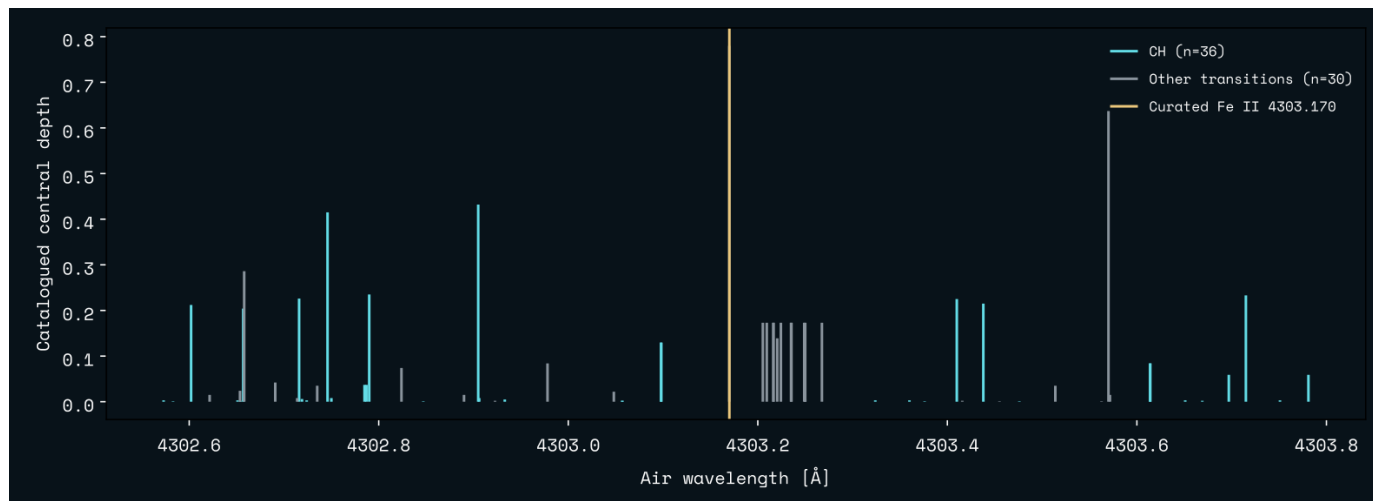
RYA-1197 Part A: the Frankenstein mean-<3D> route is HELD FROM HEADLINE publication until the Bride validates it. It inflates A(Fe) by roughly +0.10 dex against the 1D siblings and exists on VIS/KP only, so it is not a comparable measurement of the same quantity. It stays in the feed deliberately: the LTE/NLTE mean-3D PAIR is the NLTE differential on ONE atmosphere (RYA-542), which is the diagnostic it is for. Not a headline value.

Saturation characterization

Committed proof census: 120 line/holding records; 100% have observed depth > 0.70. Median observed $\log_{10}(EW[m\text{\AA}]/\lambda[\text{\AA}]) = -1.252$. In consistent Å units this is -4.252. These diagnostics characterize the audited saturated pool, not every newly added product.

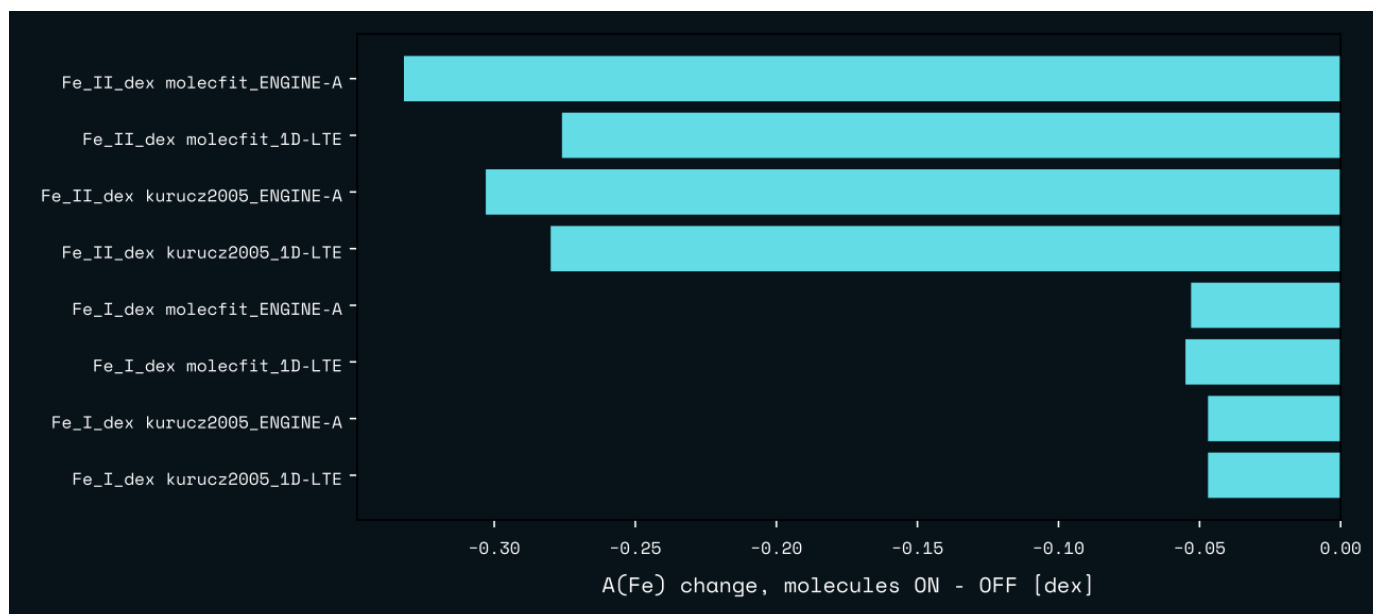
Problem-line and model diagnostics

CH G-band artifact



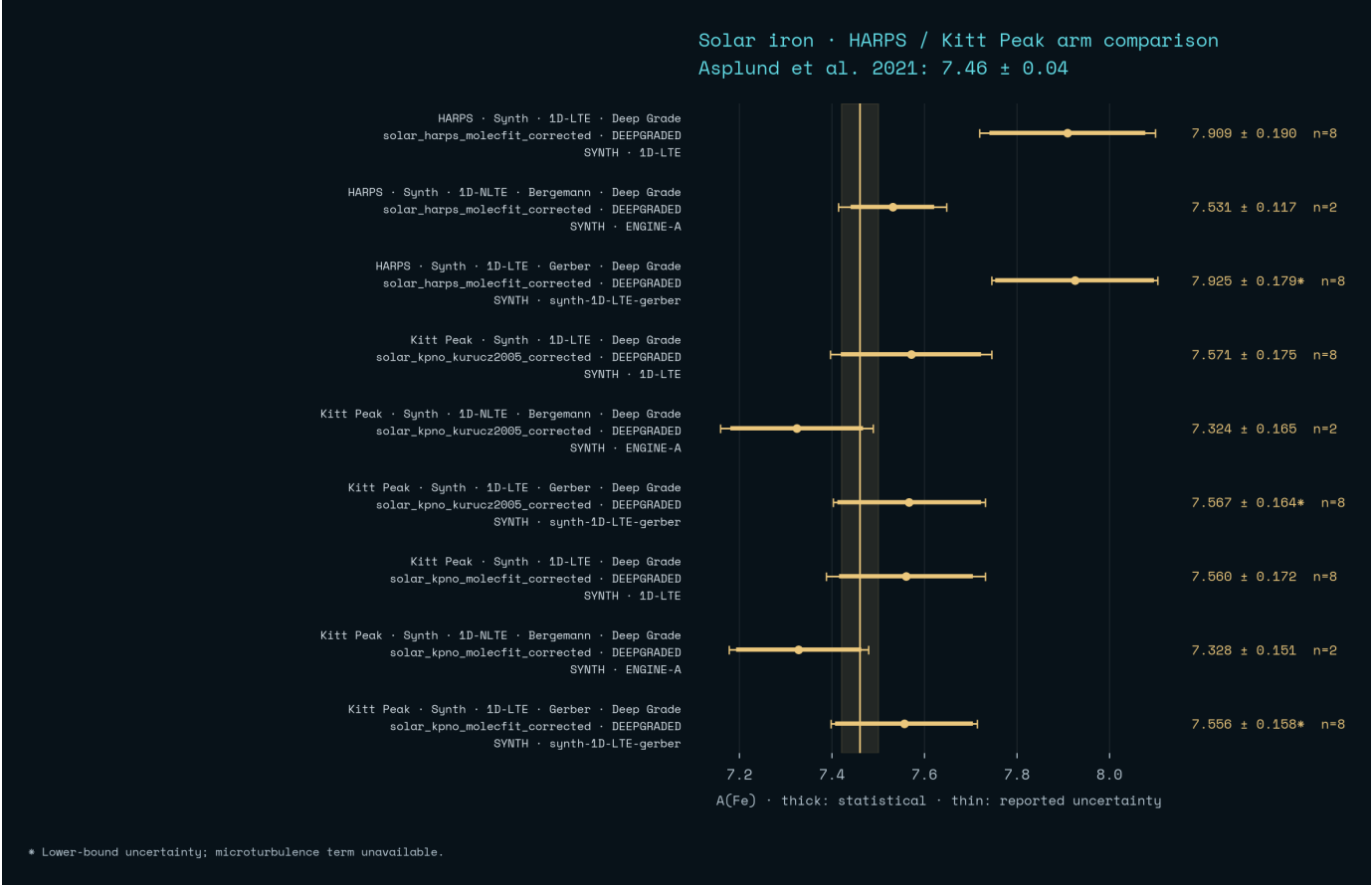
CH G-band blend artifact — the synthesis list cannot represent the dominant absorber · $A = 10.0$ (+2.5 dex). RYA-1191 blend audit: 66 catalogued transitions in the 0.62 Å fit window, top contaminant CH I at 43.8% of the summed depth against Fe II's 11.1%. The VIS synthesis list GESv6_atom_hfs_iso carries 206 species and NOT ONE molecular, so the fit had nowhere to put the CH absorption except Fe II. RYA-515 independently counted 36 of 65 transitions as CH. The plot shows the committed transition census in the proof generator's fit window; it is not a synthetic spectrum. [Source evidence](#)

Near-UV molecular opacity



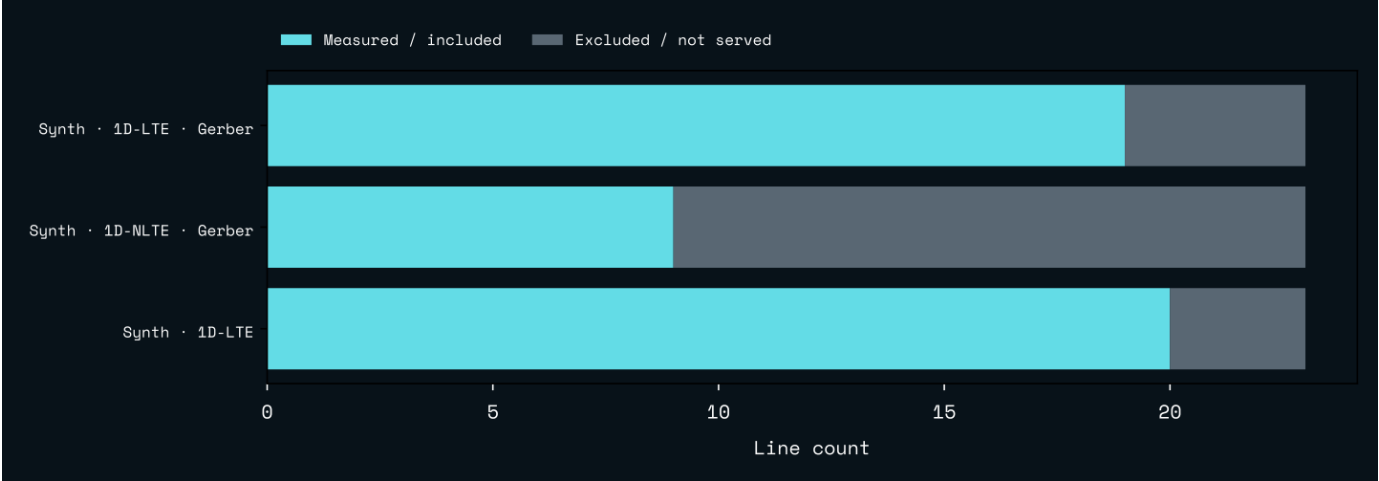
molecules OFF vs ON on THE SAME CODE and the same pool. Differencing against the published values instead would have measured the molecular lever PLUS every code change since the feed was written on Sept 3 -- which is about -0.035 dex of the total. [Source evidence](#)

HARPS arm offset



Current Fe II VIS products, separated by holding and engine. The arm offset persists after the curated CH artifact was removed. [Source evidence](#)

CRIRES+ H model coverage



Included and excluded line counts from each current product. Excluded rows are not automatically failed fits; the per-line reason distinguishes unavailable NLTE coverage. [Source evidence](#)

Download the evidence

2676 graded per-line measurements exported. Evidence for 21 graded products is missing or does not reproduce the feed; 0 exported lines lack an unambiguous gf match. The download is explicitly incomplete until those upstream records are supplied.

[Graded per-line abundances \(CSV\)](#)

[All product fields \(CSV; nested fields preserved as JSON\)](#)

[Per-product line-evidence coverage \(CSV\)](#)

[Source Fe.json](#)

Reproducibility

Generator scripts/generate_fe_publication.py v1.0.0

Science source commit 370a66e8de93b5b3453edd56114f2814a0d06bed

Fe.json v1.163 · feed timestamp 2026-09-11T04:50:42Z

Generated 2026-09-11T05:30:36+00:00

Feed SHA-256 0b49055ba5dc5931f126b2dba263c387570e6ba5df1893ef57c1bbde33442b62

[Refreshed product tracker](#) · [Build manifest](#) · [Fe I appendix](#)

References

Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). [The chemical make-up of the Sun: A 2020 vision](#) (doi: 10.1051/0004-6361/202140445). AGSS21 Solar reference and the green literature comparison; Table A.2 defines the Reference Grade Fe I line set.

Lodders, K.; Bergemann, M.; Palme, H. (2025). [Solar System Elemental Abundances from the Solar Photosphere and CI-Chondrites](#) (doi: 10.1007/s11214-025-01146-w). Table 6 present-day Solar composition, shown as the gold literature comparison.

Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). [Non-LTE radiative transfer with Turbospectrum](#) (doi: 10.1051/0004-6361/202243673). The Gerber Turbospectrum non-LTE synthesis implementation named in the model rows.

Den Hartog, E. A.; Lawler, J. E.; Sneden, C.; Cowan, J. J.; Brukhovesky, A. (2019). [Atomic Transition Probabilities for UV and Blue Lines of Fe II and Abundance Determinations in the Photospheres of the Sun and Metal-poor Star HD 84937](#) (doi: 10.3847/1538-4365/ab322e). Laboratory transition probabilities used in the resolved Fe II per-line evidence. Coverage and unresolved products are listed in the downloads.